Course Objectives: Computer Networks (Undergraduate Level)

The primary objective of this course is to provide students with a comprehensive understanding of computer networking principles, architectures, and protocols that form the foundation of modern communication systems. By the end of this course, students should be able to:

1. **Understand Fundamental Concepts**
   - Explain the basic principles, terminologies, and components of computer networks, including network topologies, models, and types.
2. **Analyze Network Architectures and Models**
   - Describe and compare different network architectures such as the OSI and TCP/IP models, highlighting their functionalities and interrelationships.
3. **Explore Data Communication Techniques**
   - Understand data transmission methods, encoding, error detection, and flow control mechanisms.
4. **Examine Network Layer Functions**
   - Analyze routing algorithms, IP addressing, and packet forwarding techniques used in the network layer.
5. **Understand Transport Layer Services**
   - Explain transport layer protocols such as TCP and UDP, including connection management, congestion control, and reliability.
6. **Explore Application Layer Protocols**
   - Identify and describe key application layer protocols like HTTP, FTP, DNS, and email protocols, and their roles in network communication.
7. **Study Network Security Fundamentals**
   - Understand basic concepts of network security, including encryption, authentication, and firewall configurations.
8. **Apply Network Design and Configuration Skills**
   - Design, configure, and troubleshoot small to medium-sized networks using simulation tools and practical lab exercises.
9. **Explore Emerging Networking Technologies**
   - Discuss current trends and future developments in networking, such as cloud computing, SDN (Software-Defined Networking), IoT, and 5G networks.
10. **Develop Analytical and Problem-Solving Skills**
    - Apply theoretical concepts to real-world networking scenarios and effectively analyze network performance and efficiency.